

ActInf Livestream #031.1 ~ "Non-equilibrium Thermodynamics and the Free Energy Principle"

Livestream | 1:55:16

hello and welcome everyone to actin flab

live stream number 31.1

yes we are live

so today it's october 19th 2021 and

we're going to be having a discussion on

the paper non-equilibrium thermodynamics

and the free energy principle in biology

so welcome to the active inference lab

everyone

we are a participatory online lab that

is communicating learning and practicing

applied active inference you can find us

at the links here on this page

this is a recorded and an archived live

stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome here and we'll be following good

video etiquette for live streams

at this short link you can see the past

and upcoming live streams here we are at

the end of october 2021 in paper number

31

and today we're going to be having an

awesome group discussion and then next

week we hope to have one or both of the

authors join us so we'll be taking a lot

of notes today and

bringing up good things to talk with

them on and all of these are

participatory live streams so if you

want to join or if you want to

co-organize with us just let us know

today in 31.1

we're just going to be learning and

discussing this paper of palacios and
colombo
and
we're just
putting out some perspectives giving
some affordances for others to join in
and
hopefully learning along the way so we
have a few questions that blue and
others have prepared
maxwell also prepared some
notes to talk about and we're just
pretty much going to
look through the questions that we have
already
written up we'll add more
anyone in the youtube live chat who
wants to ask a question they're more
than welcome to do so and we'll just hop
around wherever we need to go
we'll start off with the introductions
so we'll just say hi and
maybe something that we liked about the
paper or something that stuck with us or
something that
brought us to this conversation so
i'm daniel i'm a researcher in
california
and i think something i'm excited about
today
maybe would be to look at this
intersection of biology and
thermodynamics
and
talk a little bit about the different
ways we can approach that intersection
this paper has some formalisms but not
all the formalisms there weren't figures

but there's so many graphical intuitions
related to free energy landscapes and
some other pieces so it's just kind of
cool to think about what are the
different ways that we come to learn
about the intersections of these cool
ideas and i'll pass to stephen
hello yes stephen i'm in toronto in
canada um

pleased to be here again and i'm
interested in this paper the general
topic of ergodicity comes up and um i'm
interested to see i often see these
papers which have sort of free energy
principle focus rather than say active
inference focus so

um just to see how in that when that
tree gets shaken how that helps or
hinders the sort of more processed
theory work which is probably closer to
where i'm

uh
interested in but this is all very
interesting thanks i'll pass it over to
maxwell

hey i'm maxwell ramstead
i'm the ceo of nested minds network an
active inference ai startup and i am a
fellow at the welcome center for human
neuroimaging
ucl in london

i'm looking forward to
discussing
this paper i think there's a lot to
disagree with
and i'll try to be collegial
in my criticism
beautiful what more could we ask for

let's
start with uh
the big question and of course
either of you if you have a different
take or there's other big questions
but this big question which is in this
paper as well as the broader
literature
what fundamental ideas from physics
are utilized in free energy principle
and active inference and how
and just
what are the implications or trade-offs
associated with the use of this idea
does it lend credibility to a theory to
be grounded in physics what does it mean
to a theory for it to be grounded in a
certain physical idea does it constrain
does it prevent us from making the kinds
of conclusions that we might want to
draw about biological systems so that's
kind of the big
question do either of you have any other
sort of
big questions that this raises or
what is the alignment of like the vector
of this paper with bigger ongoing
threads in our research community
well i mean i guess one outstanding
question for me is uh what is the
horizon of relevant literature
in terms of time
course
that one needs to consider
before you know trying to
engage with and criticize this material
it's something that comes up i think
and it might be useful to discuss

maybe before we go into the specifics
there just when we're engaging
in
interdisciplinary or transdisciplinary
scholarship or we're diving into fields
where we might not have taken courses to
the undergraduate or higher level just
it's a really important question how do
we
make
appropriate claims you know how do you
take a little bit of a of a dip into the
rabbit hole
and come out with something that's
useful
rather than just being more confused
than when you made the initial search so
these are
big questions any
thoughts are just how do we determine
what literature are relevant like if we
wanted to write a paper on something
related to active inference or just in
general what do we do do we do keywords
do we only look at the last five years
are you asking the question rhetorically
or do you want us to chime in oh maybe
steven first with a raised hand but then
just either
a mute stephen and then yes
stephen unmute and then continue
i'm probably not the best person because
i keep digging and digging and digging
but um
i think it's a good question i i i'm
from a chemistry background a long way
back and i always find it interesting as
well because you know a lot of stuff

that's
used is or developed it comes from
chemistry actually um
uh and uses physics but uh
you know the the the like you say that
we're in this kind of
need to
approximate and an approximation what
like there's so many things that come up
that are like new
new
ways of describing stuff which makes
total sense like i i've read through the
free energy principle and kristin's work
about the talk about approximation
science and that science is is
ultimately an approximation
and then you know it takes me a couple
of days to then get my head around that
and it's like yeah that totally makes
sense but the implications of all of
that are huge the implications of being
in non-equilibrium
states
itself
disrupts a lot of the accuracy of
of
certain ways that you can roll out
things from physics and are you rolling
out into chemistry which is normally
equilibrium chemistry
are we rolling it out straight into
biology are we rolling it out into
something slightly different with all
this modelling so
i think it does
cause some challenges in terms of
discussions because

it's easy for other people to start
telling
the free energy principle
and active influence community what they
are and then start telling them
and framing what they're not
and by which point it's kind of hard to
have a discussion because it's like the
straw man's been set up
and i think that can be a little bit
challenging for some conversations to
move forward because now you've got a
complex idea
trying to
first of all
counter
the fact that other people's paradigms
have been challenged
uh so i think that itself
um i don't know if that's a literature
thing or whether that's a cognitive
thing but i think that that is part of
the challenge that seems to be faced
agree steven thanks for that and it's
maybe a good segue maxwell we can just
start to hear some of your thoughts and
all be typing some notes and maybe
asking some
follow-ups and unpacking because i know
you'll have a lot to
say on this so maybe just
go for it and i'll just type along and
we'll see where we go i mean um i guess
that
probably the thing that i find most
puzzling about the paper is that it
claims to be the first in the literature
essentially

to be looking at the connection between
the physics of the free energy both
physics
and the free energy principle
and
i find this puzzling because it seems to
me that
the connection between physics and the
free energy principle has been really
maybe thus the focus
of active research in our community for
like
four years or so um so i mean you know
the
carl published or at least pre-printed
his uh
particular physics monograph in january
2019
um and you know prior to that we had all
been working on
i mean effectively the connections to
physics so
uh i find this problematic because the
work that we're going to discuss today
focuses on you know
older formulations of the free energy
principle where the connections to
physics were maybe less obvious
i mean as most of us in the community
know now
the formal foundations of the free
energy principle have been subject to
extensive debate over the last you know
three years or so
uh
kind of martin beale and colleagues kind
of started um started us off in this
direction and you know uh since um

martin and uh you know a few other uh
groups of uh produced
uh you know
have stimulated interesting discussion
through their very interesting and
technically rigorous critiques of the
free energy principle but really it
seems to me that this has been like
a focus of our discussions for the last
several years the connections to physics
and so on um
you know and where this has gotten us is
uh the bayesian mechanics for stationary
processes paper that was recently
released
um which accounts for
the critical
remarks that have been raised and you
know it so really where we're at now is
that we have uh formulations that start
from
non-equilibrium statistical mechanics
and information geometry
um and end up
you know saying well
all systems that have stationary density
dynamics and a markov blanket
will look as if their internal states
are uh
you know modeling in a
loose sense or in a more technical sense
encoding the parameters of probability
distributions or bayesian beliefs about
external states that's sort of where
we're left with and you get you get to
this from the physics so i just found
that the um
the remark was a bit puzzling and the

choice of literature was also i think a little

uh

puzzling for the

the same reasons i mean i i have more

specific thoughts but

uh

i mean you know this is sort of for your

first bullet point in response to the

last one that we added what's the

relevant time horizon i feel like a

rapidly developing theory like the

active inference framework and the free

energy principle i mean if you're going

to be producing

technical critical material

you

should probably be working from the most

recent

versions i mean i find it puzzling for

instance that the authors spend so much

time talking about ergodicity

when the word ergotic doesn't

appear

or just just about never appears in the

particular physics monograph so anyway i

just find that there's a kind of

disconnect between what's presented here

and the actual work on the physics of

the free energy principle

that

that we have seen and uh

you know i think that leads to

several of the arguments uh being

proposed

uh being a bit um

missing their mark to put it

diplomatically

thank you for the concise summary
maxwell so steven with a raised hand and
then we'll continue
could i ask a little question as well
that relates to this is in the very
first
part of the abstract it talks about the
um
inevitable and emergent property of any
ergodic random dynamical system at
non-equilibrium steady state that
possesses a markov blanket so that's a
quote from the paper from 2013.
the thing is it says
before that according to the free energy
principle life is an
in and then it quotes inevitable and
emerging property which is the quote but
i'm not sure
that life is ever talked of as an
inevitable and emerging property of any
rand ergotic random dynamical system
in
terms of how it's referenced and i'm
wondering if there's a bit of a
confusion in the paper there where
um
it talks about this that quote
but makes it as if that's what life
that life is an emergent and an emergent
property and i think it's
um i think what we're talking about is
um
adaptive reduction of surprise
um or bayesian um inference
properties is an inevitable and emerging
property but i don't know if that would
be life but i don't know what your

thoughts are on that or whether i'm
misreading that
that reminds me of one of my favorite
papers what is life answering
schrodinger's question by maxwell at all
in 2018
um because
maybe you could even say something to
this what is the relationship between
reducing surprise and life
and do these recent technical
developments in fep
help bridge the gap between inanimate
and living
cognitive or computational processes and
where does physics play into that
that's that's very interesting i mean i
think the
the fep itself
uh
is not
a theory of life just in and of itself
right what is what it is
it's a theory to understand
uh
systems that are able to maintain
themselves in non-equilibrium regimes
effectively uh
so or i mean more precisely you know
stationary systems with a markov blanket
which is to say the same according to
the theory um
so i've gone on the record as saying
that the free energy principle can allow
us to construct
a theory of life
but the claim isn't just that the free
energy principle in and of itself

allows you to do that
i think you know
the fep needs to appeal to auxiliary
hypotheses from you know biological and
other sciences
to really do that i mean it's a theory
of thing-ness
in
physical systems
that interest us
and a subclass of those systems
are biological systems
um so i mean it is an aid in the
formulation
and probably the foundation of a
formulation of uh living systems so so
long as we agree that living systems are
a subclass of persisting systems and i
mean there's some talk in this paper
that maybe this persists
living systems of persists might not
be the right way to think about life but
yeah i think it acts as a foundation
i mean but then you need to add a lot
more
to turn that into a theory of life i
mean if you read my schrodinger paper
carefully for example what we propose is
a variational approach to neuropathology
uh we're not saying that
the free energy principle in and of
itself can just offer this like the free
energy principle has to use as it has to
be used as a foundation from which you
can build
these uh variational approaches to
neuroethology to niche construction to
ecology or what have you

um

so i mean but you know that that might

just be uh picking it nitpicking so like

i uh you know

well i don't want to double down on that

too much

it's

good points and it's almost

related to the definition or the

distinction between like a discipline

and a trans discipline

if

fep is a discipline versus if it's

connecting other disciplines variational

neuroethology it's right there in the

title we're connecting across different

areas

and the framework of thinkness

is one of those enabling features

that helps it be an integrative theory

but potentially it doesn't fulfill the

disciplinary

um uses

of theory so that's one interesting idea

um i guess there's many other

ways to go or things to ask i guess i'm

just i'm curious again like maxwell what

do you think in the last few years

how have we clarified or learned more

about where active inference and fep are

related to each other

so how was that thought of in

five years ago maybe when you were in

grad school

and then what do you think is sort of

contemporary greetings blue

in terms of the distinction between

active inference and fep and then

welcome blue and you can add anything
you'd like
and i know some people want to mark the
difference between the two i think
personally i think it's more of a
sociological thing
uh because
no one or of just about no one is
contesting the validity of active
inference models of cognitive
systems but there's a lot of pressure
from different circles on the free
energy formulation so i mean it might be
a sociological thing where it's just
easier to say oh i don't work on this
more you know
metaphysically sounding thing i just
work on computational neuroscience and
that's straightforward uh i think
there's no warrant for that from a
mathematical point of view you should
just
for example look at the way that things
are derived in the bayesian mechanics
paper
um
yeah i i feel like uh active inference
just falls out of
the the core bayesian mechanics
i mean i'd recommend everyone read
through that bayesian mechanics paper
it's really fantastic
it's very clear
one of the big differences between it
and its previous free energy papers is
that uh it's authored by
or at least there are two authors out of
the four on the paper who are proper

mathematicians working on you know the
mathematics of non-equilibrium
statistical mechanics effectively so uh
you know that paper really uh i think
meets the standards of rigor in
mathematics you know you have
an existence proof
for the synchronization manifold that's
sort of at the core of the whole formal
construct the thing that links internal
and external states
and then you have a construction of it i
mean it's very straightforward and
having having established
uh
you know the kind of core
formal
apparatus it's unfolding and by the end
you know you what you do is ask kind of
well all right let's split the blanket
states up into sensory and active states
and active inference just falls out of
you know
making the
active states subject to
the dynamics that those are unfolded
throughout the paper so in my view the
connection is immediate
um
yeah
uh i i i will grant that you know
they're
some of the computational models that
we're using are a bit more um
they look a lot more like just vanilla
you know computational neuroscience
models uh
you know their palm dps and stuff like

that but

but what the math that underwrites it

all is the math of the free energy

principle so i don't really

i don't really think the two can be

separated

thanks maxwell and um the basic

mechanics paper we discussed in

live stream number 26 so

it's um it is a lot to keep up with

sometimes but that was a very

fascinating discussion so blue and then

anyone else

i was wondering if that was bayesian

mechanics for stationary processes the

one we did yeah that's precisely it yeah

um so i sorry i'm late on the scene but

um i am curious about whether is the fep

a theory of life or does it allow us to

construct one and does it only apply to

life or can we take it in this kind of

pan psychic direction

um

where even molecules like want to

minimize their free energy and

self-assemble to become structures

that's a very good question um

i would say that the free energy

principle is not a metaphysics

um

so

it's it's a physics

so it's a formal framework

that

uh

may or may not give us

grip on the systems that we're

interested in

so i mean you know i and
i i know there is some philosophical
literature that presents it is more like
you know a metaphysical statement about
what life is
but i think it's more useful to think
about the bayesian mechanics as just an
extension of mechanics effectively you
know the
the same way that you know statistical
mechanics and quantum mechanics
are flavors of mechanics the bayesian
mechanics is also
you know and i realize that it you know
it's a little deflationary
uh it's a more instrumentalist reading
of the free energy principle but it's
not completely deflationary in the sense
that it the model says
that the best model
of livings and other existing systems is
as a model of their environment
so
you know
there's a kind of double instrumentalism
at play i think in in the free energy
stuff
to answer your question a bit more
directly i think it is it can be the fep
can be used to construct the theory of
life
in and of itself it's a theory of
thing-ness
um
but yeah it's a great set of questions
to discuss and i mean this is only my
view like i don't mean to
settle this once and for all there are

people who want to defend a more um
metaphysical
view on this
um i guess the question is
how far do you want to take the
commitments
uh of the free energy principle i mean
you know the there are kind of two
irreducible assumptions that go into the
math there's stationarity
which by the way is not the same as
ergodicity
so you know uh i think a whole section
of this argument just doesn't apply at
all anymore
the one presented by colombo and
palacios
and markov blankets
so i mean you know if if you think that
uh
you know systems that exist at some
scale have to be stationary
and you think that lit you know systems
that exist really do have a markov
blanket
then you can be realist about the
assumptions
but i kind of feel that it works whether
or not you take the additional
philosophical step of committing
hardcore
to the realist assumptions
uh
you know it works just
it works as an you know
an as if
theory if you will i mean the whole
thing tells you like living sorry uh

systems that uh
existed on the at non-equilibrium study
stated have and have a mark all blanket
will look as if
they're inferring
external states
and i mean um
yeah it i think this this looks as if
stuff is really
important
and i realized that you know for for
some of us it may be a little bit
deflationary uh
but i don't know if more is warranted
from the math i think that you know
there's a
yeah we just need to be kind of careful
um
thanks for identifying those two key
pieces and maybe we'll have more to
think about with the stationarity and
ergodicity so blue with the raised hand
and then stephen
so just a couple things like when i say
pan psychicism i wasn't trying to go
into like the metaphysical realm but
because it's physics because it's not
metaphysics can it then apply to life
and non-life like can we just go all the
way down yeah and so in that kind of way
i wasn't really implying as a
metaphysical construct but more in like
the direct does it broadly apply to
non-life as well as life yeah and and
then also um just a couple more things
so
systems have a markov blanket like
maxwell that's you being entirely

realist

like i don't have a markup blanket but
can i be modeled with a markov blanket
probably um so so i don't i don't know
like you know i don't curl up in my
markov blanket at night when i go to bed
comfy

and then just lastly like going back to
the paper um this like so i don't know
that was one of the main questions that
i have for the authors is this concept
of they say like systems are in dynamic
equilibrium

like

i don't think that the non-equilibrium
steady state density is the same as
dynamic equilibrium at all and that was
one of the main questions that i raised
and like dynamic equilibrium really is
not it's a chemical process like of a
reversible chemical reaction so i i
don't i i really kind of missed that
that was one of the key things in the
paper that i just didn't understand at
all yeah i uh i mean i

i very much agree with you on the points
that you're saying

uh

the requests that you're raising here um
so i mean regarding the pan psychism
thing i mean yeah you can from physics
now kind of say that any system that
exists kind of looks as if it has a mind
uh you know then you might not want to
go full on

the metaphysics of pan psychism but it
certainly warrants that kind of claim i
mean in my view um

this paper is making uh a few critical mistakes uh
and this is echoed a lot in
um you know a lot of the literature coming from more like
four e people writing about this i'm just gonna cite you know two sentences from a paper pre-printed this year
uh i won't i won't name names but you'll see the style of what i mean to be as fair as possible we'll do this as in
criticize the fep by attending to the more recent published versions and variants of these claims and then the next section starts literally the next sentence the work presented in first in 2013
is central to practically all the proposals comparing the fep and in action and it just goes on and stuff and it's
i i sort of feel like yeah the physics foundation has moved forward like you know blue you're pointing it out it's all about non-equilibrium steady states
and
non-equilibrium steady-state dynamics and you know stationarity that's importantly different from ergodicity uh you know the one of the nice things about the the bayesian mechanics paper
is that it it abstracts away from those assumptions i mean to be clear the free energy principle at its core never really required that the systems that your study be ergotic

we assume an ergodic density in some of the earlier publications because it allows us to finesse one step in the derivation

substituting

the time average for the ensemble average and one of the really core derivations

but you know as uh carl himself had been pointing out for a long time

you could finesse the math into you know something else like there's no reason

why you have to stick with

the ergodic assumption it was used to finesse

one step of the derivation

and you know what lance de costa

and a

company have done in this bayesian mechanics paper with carlos precisely to do the laborious work

of actually generalizing

the math to

non-ergotic systems so i mean i think

that's really important the bayesian

mechanics paper um

and you know we're in discussions with

the people active discussions of the

people who've produced the critical

material and i mean at this point i

think everyone

essentially agrees that lance's new

derivations respond to

uh

you know the the critiques by aguirre

and

colleagues that came out earlier this

year

and also the beel at all critique

remember the the core of the beel at all

critique is that uh

for ou processes

in general

of the sort that carl was using you

can't assume that

the synchronization manifold will exist

between internal and external states

in fact what martin and company have

showed is the for a generic ou process

you can't just you it does not exist

um but you know that

that doesn't mean that you can't find

very broad conditions under which the

synchronization manifold exists and

that's what

lance and company have done in this new

paper

uh

aguirre's critique aguero at all this is

the sussex group also did a very

interesting analysis of the math as it

figured in the uh

particular physics paper

uh

you know showing that actually the the

precise assumptions that carl uses in

the paper would restrict the scope of

the fep to a rather narrow class of

systems but again lance just rederived

all of the equations so the formalism is

now effectively different and what you

can really see from the math is that it

applies to

non-linear systems and to non-ergodic

systems

and this leads me to another point um i

think the the the assumption of
non-equilibrium steady state is often
not understood in the way that it should
so non-equilibrium steady state is a
characterization of the density dynamics
it is not a characterization of the
system itself if i if i say that a
system is at non-equilibrium steady
state i'm not saying that it is steadily
in some
state you know that it just that it
occupied what i'm saying
is that the the probability landscape
that underwrites its dynamics is
stationary
or has stopped evolving
um those are two very very very
different claims so it doesn't follow
from the fact that a system has a nest
density that it is history-less it
doesn't follow from
uh
you know the the existence of the next
density that uh there should only be one
attractor like you can have
metastability
in an fep system you can have
non-linearity in an fep system you can
have non-ergodicity
in an fep system
um
so
yeah i mean i i think
you know
that means you know that one of the
three
sections the critical sections in this
paper are just completely off the mark

and you know
there's this new paper
by some of the tenors of the inactive
approach
on precisely this kind of stuff as well
and
unfortunately the same mistakes are made
um you know ergodicity in in the context
of the fep
uh
i mean really just means that the the
the quantities that you're uh
observing or measurable
um it means that you know all the
average of your measures converges
on average to a measure
of the average
you know or of the mean that you're
actually trying to
calculate in a heuristic sense
so it never meant that
systems don't have a history it never
meant that systems necessarily converge
on one attracting
set
and so i i really feel that that kind of
takes the teeth out of
most of the you know formal criticism in
this paper uh the other part of the the
other critical parts of this paper talk
about you know the challenges of
modeling
biological systems using state spaces
but that
those sections to me don't amount to
much more than just saying it would be
very difficult to do this
um

therefore

it's not doable or something like that
and you know i i i have a lot of friends
who are computational biologists and i
speak to a lot of people doing this and
it's sort of like well

you know some work would have to be done
to write a paper to respond to this but
you know what's the expression one
person's uh modus tollens is another
person's modest modus ponens or
like you know i sort of feel like all
right well you know you say this is
impossible uh i know that formally you
know alex alec chan's

is a really cool guy and uh
he's brilliant he and the sussex team uh
have recently

recently i mean this is two years old so
it's not that recently and i feel like
this should be

accounted for in this paper but have
recently automated uh state-space
construction

um in the free energy framework like we
have active inference models now that
can kind of automatically assemble
and can handle up to it's either 10 000
or a million

states

so you know

that's a lot of modeling power and i i
find the example that they discuss in a
footnote there

uh you know however this is on page 11
however the size of the genotype space
for a protein consisting of n residues
chosen from an alphabet of amino acids

the size m is m to the n and then they say oh well you know it's like 20 to the 100 or 10 to the 130 possible

uh states if you wanted to model like you know amino acids what they're not considering is the nested formulation and i i kind of feel like nested nest densities should have been at the core of this argument and figure nowhere

uh yeah i mean of course if you think that you can just plot all of the states of the system across all the scales at which it exists in one you know homogeneous state space you're going to run into a problem but you're not if you carefully construct these state spaces in a kind of nested fashion um you know like a friend of mine van the tower

uh works on

i pulled this paper up he has a really cool pre-print a deep learning approach to capture the essence of candida albicans morphologies and i mean basically there what they do is start with a population of candida which is a fungus for those of you who don't know

and then you you randomly generate a crispr and you introduce it into the population and you see how the population develops

and you plot each populational trajectory in a 30 000 parameter hyperparameter space effectively and you see the way that like the candida self-organize into pockets depending on

their you know the kind of crispers that
you put into them uh

i mean this is not like fringe work this
is just routine computational biology
where we're already you know plotting
phenotypic expression in 30 000
dimensional
spaces

uh you know

this isn't like forthcoming

work to be done i mean it's really just
standard work it's interesting how
you know i i think they they modified
the paper we had discussed this in a
live stream before and they

they

nicely added some uh responses to some
of the criticisms that i i raised in
discussion

um

i'd have to

pull it out but you know i i said well
you know that we're using this stuff in
studies of uh cancer dynamics and you
know like so uh i i wrote in a paper you
know

we can think of a biological system's
extended phenotype as the set of
extracting

attracting states of a coupled dynamical
system and this is supported by studies
of cancer genesis and propagation

uh i can link the papers

and you know they say well you know

these studies provide evidence that

cancer can be fruitfully modeled and

understood as an attractor in a

dynamical system but they don't provide

us with evidence that all biological systems occupy an attractor and i'm sort of thinking like well no but it you know this is a counter example to you're saying that it's not doable in a real biological system you know uh so i you know if i if i'm being snarky i feel like this is armchair biology of the worst kind kind of you're telling me that what i'm doing is impossible while i'm doing it and you know and you're doing it by appealing to like you know an old conception of the physics uh by talking about assumptions that aren't even part of the formalism anymore i mean i i just uh i find that there's a lot to object to in this paper both in terms of the substance and in terms of the general kind of strategy of kind of gotcha in uh now does the free energy principle need to be discussed critically absolutely and you know i i should have caveated all this by saying that some of the most interesting and stimulating uh intellectual exchanges that i've had in my life have been with you know the people creating critical material uh but you know the the people that are generating this

interesting critical material are
actually engaging with
the theory at its formal core
and
you know i i sort of feel like some of
these things would have just been
cleared up with
you know a few conversations and looking
at the
more recent material but
now i feel like i'm just kvetching
at this point
thank you maxwell that's what these
discussions are
for and hopefully bringing into the open
it's just a multi-perspective
scenario
to give you like a specific idea why
this is problematic like if if you focus
on ergodicity and not on the nested nest
densities then you're going to say
things like well first you have to show
that there's an attractor and then you
have to show that the relaxation time is
biologically plausible and they claim
that neither of these things are done in
the free energy literature when
literally
you know the core 40 50 pages of carl's
physics monograph deal with and you know
dan uh i i i know you we've we've read
it all together during the last year
like
i i don't know how you how they could
have missed this i mean it's it's the
physics monograph this is supposed to be
their topic the core of the physics
monograph is all about you know uh

information length
uh effectively right and relaxation
times back to non-equilibrium
steady-state and how all these things
kind of work in a nested way to get you
biological dynamics so it seems to me
like they are just completely missing
their target
but both by you know working out
on
on formulations that are out of date and
that introduce assumptions that are no
longer at play
and and also by just really ignoring
you know
the physics stuff the free energy
physics stuff that they claim to be
criticizing uh
you know
so
we need to be um just like if we were
going to catch a baseball we need to be
performing anticipatory
knowledge work not merely reactive
especially to previous uh formats and
then or if it's reactive i mean you know
react to the recent stuff i mean you
know there's been a concerted effort
to
ground this
in the physics of non-equilibrium
systems like non-equilibrium statistical
mechanics information geometry i mean
if you're going to be writing a
criticism of that stuff
then you should probably be writing
about that stuff is sort of my point
and you know and by the way this often

goes really badly when i point out that
hey you know you're working from a
you know an outdated
thing they say oh you know
the free energy version 25 from this
morning and blah blah but i kind of feel
like you know
my friend if you want to write a
criticism of something
like you should be as charitable as
possible with the target of your
criticism on the one hand and you should
definitely not be working
on versions of this that no one in the
core community is still working on and
there is a dimension of like if you
can't keep up then i don't know you know
i don't know who benefits from this it's
sort of like my
my
questioning it's like
okay so steven and then blue
yeah i think
thanks for the giving us a good good
insight there i i think the the site the
sociological psychological piece
is is present in all of this as you say
and i think i've mentioned this before
to you maxwell actually is the
the danger of systems talking about
systems i think even if you were talking
to people in the international society
of system sciences they often talk about
systems system dynamics
and everything else things complex
systems kind of gets lumped into
complexity right so there's a danger
because i don't think

carl necessarily would say fep system
right and that term system i think gets
you into a lot of challenges because the
folk psychology of what a system is is
what we see it as from the outside
and you know these non-equilibrium
steady states i'm wondering whether we
say mark he says blankets all the way up
and down

but i'm i'm not sure if it's more like
thing this all the way up and down is
safer than systems all the way up and
down because

things can have all sorts of you know
and they can have states and they can be
what they are

and the question really becomes then
within all that ergodicity which unlike
physics which tends to be about getting
things to a certain number of decimal
points of accuracy jen

it's a messy thing you know i was
talking about that thing about
approximation science i mean again that
gets your head around what that that
makes suddenly you got well the main
thing is you just need enough
of whatever this stationary
extractable

data is to be able to do something with
it and i'm kind of thinking that maybe
because it in the same way that in
chemistry if you have a beaker
and you stir it up
and basically the thing will have its
dynamics it will move
based on the energy between the
different things and basically entropy

is just basically the
the force that's pushing it in the way
that you don't expect it to it's
basically
you know we know it does this from this
equilibrium state we know that and
entropy is just the
thing in the middle now no one has a
bloody clue what's why
that's just a black box right now but
that's because they're going from now
when you go to these
non-equilibrium
far from equilibrium chemistry state and
they're held
which you don't do when you mix up a
beaker and just go from a to b now we
don't know how many
non-equilibrium states
these reactions go through to be honest
in those cases now
but you wouldn't call the beaker a
system you call it
a solution you'd call it a you know you
might call the vessel and the chemical
plant in which it reacts a system but
you just sort it kind of a beaker of
stuff or things that's doing stuff and
things right and you sort of see what it
does and then it's like what can we do
that's useful
and normally what's useful is what
energy shifts but in this case you've
also saying the entropy can be useful
but not in the way most people would
the what we could say the folk systems
thinkers
would kind of tend to think about it so

i'm just wondering if there's a danger
of
using the word system
yeah i i would agree i mean uh
i think we use the term system just
because uh you know all of this grows
out of dynamic systems theory in some
sense
uh you know
carl talks with particles
right like uh
a a system with a markov blanket
and does the free energy stuff is a
particle in his account um so that yeah
there is a uh
so there's a there's a linguistic
danger here because i agree with you
this is not what people think of when
they hear the word system a lot of our
exchanges as you pointed out stephen
have turned around that specific point i
think the the term is important because
especially in the context of relating
the fep to like the four e approaches
i think it's not
appreciated sufficiently
that the fep math is what happens if you
take dynamical systems theory seriously
um you know so when i was doing my
undergrad i was reading the four e guys
and
the fourier people
because it's not just guys
and
uh yeah the the fourier people were
telling me all right well you know
computer theory you only get so far in
the study of cognition you have to do

dynamical systems
learn dst math and learn the physics of
complex systems so i did
and several of us did
but what happens is then if you continue
asking questions along that direction
things arise like well how do you
measure distances between states
and if you start asking that kind of
question inevitably
information and entropy creep back into
the picture
because that's how you measure distances
between
states effectively
is in gnats
and other information theoretic units so
i mean this is to me the great irony of
this whole you know fpp versus an
activism confrontation which is slowly
turning into like one of the big
discussions in contemporary philosophy
is i don't think that the
fourie people fully appreciate that the
fep stuff is just a prolongation of
their thinking style that if you
if you really want to take seriously the
core tenets of embodied cognition then
inevitably you are led back to
information
there's no escape
it's really if you want to take their
own framework seriously
there's no other place to go but here i
unfortunately have to uh leave now thank
you maxwell thank you for listening to
me rant about this i'll try to be there
next week um and yeah i mean uh keep

kicking ass i mean dan blue
uh i i just love active inference lab
and i'm so grateful for the awesome work
that you do keeping this community alive
stephen it's always a pleasure to talk
with you and i'm glad to see that you
know there's uh so much uh
so much like yeah you're just there all
the time and i just love your uh you
know your
your contributions to the conversation
it's always a pleasure to talk to you so
a big love to everyone uh
and i will be uh in touch shortly uh
apologies for the snarkiness it's just
you know
this is the kind of paper that really
grinds my gear
i thought it was it was uh good to just
point out a few of what i i think are
the issues
thank you maxwell see you soon
goodbye
this uh blue will go to you but this is
uh you know it's a key difference
between being an organizer and a
facilitator or a participant
um
we're
here to help scaffold
individuals to participate and their
perspectives
and uh that's the second order
cybernetics so things will get more
contentious
than we can even imagine
and someone just has to hold down the
broadcasting

make sure that hands are being raised
and that protocol is being respected
because there are serious first level
questions to resolve
but we can have precision in our second
level policy and that's gonna return to
a question in the live chat but first
blue with a raised hand thank you
so i just wanted to um you know say that
a lot of what both maxwell was saying
and um what stephen said just really
underscores the need for
consistent definitions like how do we
define a system and also like a
consistent synthesis of the fep like
historically
where did it start where has it been
where is it going
and like what where are the landmarks
like for example this basin
bayesian mechanics for stationary
processes paper you know has revamped
the math and physics and when we're
making a new paper or a new argument are
we working with
the current version of the model
absolutely nice uh foreshadowing of some
work that we're involved in but if
anyone is interested in sort of
scholarship and a history from an
empirical perspective
on fep and active they should get in
touch with blue or i because this is a
project that we're super keen on
but it's a very important point and
that's why in our dot zero we had like
this slide
with um the active inference lab the

working ontology so we're going in a
participatory way
and we're working through these terms
and we're finding where it's quoted in
the literature how it's used in the
literature
and then we're adding notes and relevant
resources and we'll be developing
curriculum
that helps make simple declarative uses
of these terms so that we don't get lost
and we'll have the terms and we'll have
the specifics of literature and it'll be
an awesome environment for learning
applying and communicating active
inference so before um stephen you want
to go with a question and then we'll go
to the live chat
well i just add to that point you just
made is i think
it is really going to be important
because we're seeing the confusions that
can happen but also it's it's likely
that there's going to be
a kind of a multi ontology
there might be something
in this whole field that um speaks to
how this changes just in the way that
um
carl fristen talks about when something
is regimes to an infomax
focus versus a hamilton
lease action focus versus uh you know
a reward
focus so
there may well be
um
this commonality and there may also be

a way to process

when things

systems

particles

become

you know derivations and you know that's

going to be interesting because there

may be some nexus

whether it's called metaphysics or

whether it's it's something more

prof

you know less less out there maybe it

can be grounded and i think that that's

starting to come up as we we hit

you know that's transdisciplinarity for

you right it's gonna you're gonna want

to know what's in the discipline piece

so maybe there will be stuff which is

kind of grounded in the biology world

and grounded in the physics world and is

going to talk

as best it can their language right and

then there's going to be stuff which is

in the applied world which ultimately

starts to go transdisciplinary at some

point as soon as you get out and get to

any sort of scale

um

so i think that's going to be very very

interesting awesome and both steven and

blue and many others join us weekly in

edu

where we talk about this so let me read

a few of the awesome comments from the

live chat and ask a question from the

live chat and anyone else can continue

to ask a question

so there's a lot of discussion on the

systems

question and yvonne

said maxwell subscribe to our

systems thinking course we will discuss

what a system is from the standpoint of

systems engineering so that's really

important thanks yvonne for sharing that

and indeed two of the fundamental

pillars or ontologies or trans

disciplines of active inference lab are

active inference and systems engineering

so those are the perspectives that we

draw on and seek synthesis and

application of

and we're all learning a lot and it will

be updating our generative model in a

way that changes our policy selection to

be able to use words like system without

going down the philosophy rabbit hole

every time and just having a way that we

can talk about systems of interest like

in dot coms our system of interest is

the live stream so how do we take a

role-based approach that embodies the

best of our values

and uses the term system and we also get

all these other pieces too so

good um thanks for sharing that yvonne

dave shared a lot of interesting

information about life and about some of

the history of a few different ideas

related to conceptualizations of life

but the question that i wanted to get to

was from michael nyc who wrote

how would you model this conversation

using this model

so that's a question we can uh think

about and i it speaks to almost like

applying because online teams are
certainly an application that's
important for different kinds of
modeling
and i'll give a first thought on this so
one thing is when you said how do we
model it
the question is beautifully phrased
because it doesn't say well what does
the theory say that this conversation is
that's setting up for an absolute
hegemonic realist answer
how would you model it
is going to be inherently
different perspectives because different
individuals and different years are
going to have access to different tools
and ideas and people so they're going to
model this conversation differently so
that's the first piece and that's moving
from that sort of absolutist modest
realism
into
more of a multi-perspectival
system
now
what are some ways that one could model
this
there could be a cyber physical
depiction where one is very interested
as we were in our 2020 voyatkin at all
paper
in the information flows from a
computational perspective like sense is
the incoming states and actions are the
outcoming states so in the context of a
video conversation
there's an audio visual feed in and

there's an audio visual going out
so we could use blankets and interfaces
and sense and action states
to talk about
just the cyber physical dimensions of
the conversation
another approach would be the
hermeneutic or computational
conversationalist approach which dave
has connected back to
a long lineage of work related to
1970s conversation theory and he wrote
it provides a micro theory for nested
mutual models focusing on the evolution
and incorporation or rejection of
predicates and their syntactic
occurrence
active inference work provides a
corresponding needed macro theory
focusing on secular evolution of the
interchanges
so one could have a communication
oriented model that focuses on the
semantics of the conversation
one could make also as discussed that
kind of a cyber physical very
computational model
and there's other types of models that
can be made and then there's a million
different implementations so that's just
the first pass but it's a really
important question and it helps us
connect
the abstractions and the framework to
the minute particulars which are the
system specific realizations
that ground us in transdisciplinarity
and application so stephen and then any

other questions from the live chat or
blue

yeah i think they're quite useful

questions those because they're

that question about how can i um i mean

it ties into a couple of things

my first interest would be say okay why

would i so there's some but the other

thing is how can i i how can i

implicitly also

link to

is available to me

what information is it

possible

to interact with you know that idea of

this

so we can bring in we

this modeling process sort of ties us in

this sort of mid ground between

how we actually want to talk about all

of this which i think is where the um

systems engineering is really useful

because ultimately you kind of have to

bring it into a system of

externalize knowledge which can be then

communicated and moved around as in the

papers and other things that

can be done

but it's also how can you start to

understand say

the process is going on in the moment so

it could be measuring what we're doing

in this real time process and someone

tracking that it could be a case of

video in it and watching back the video

with an interpretation thereof it could

also be you know if we were wearing

sensors you know they could be just

trying to model the dynamics of us as
things you know there's a just trying to
say
how could i know
what i can even trying to
feed into a system so i mean that one of
the classic cases that's recently there
with ryan smith
their work on interception they had to
devise
a method for recording
first they did it with the heart and
they changed the way the heart was
measured but they also did some
recordings of the stomach and the
stomach region
they had to find a way to get data
to find out what could then be modeled
and
was that a system they were modeling or
was it almost the dynamics they were
modeling and of the things you know and
i think that's i think and just seeing
how all the dynamics change over time
and i think this is this is where maybe
there's a lot of challenges um that's
going to be faced because we've had this
conversation in this modeling world is
you know
jumping things into a system engineering
world which isn't
only a non-equilibrium philosophical
world it's quite useful for applied work
because it does that ability to make
something a system
in a way that's intentionally that and
it has system of interest just like
daniel was saying

and then you've got
but within that our system of interest
is about going beyond
the systems of obvious
presentation you know there's stuff
going on
under the hood
which is not necessarily um
intuitively available at all so i don't
know what your thoughts on that daniel
but i think it's interesting
yeah the only short thing i'd say is
like even if the system of interest for
example is the mind of someone being
educated for example in the curriculum
of active inference lab or the audience
who we're speaking to is our system of
interest
or even live stream
having your system of interest be the
learner's mind or brain
doesn't mean you're looking for the
neuron level description
having the system of interest to be the
live stream or the audience doesn't mean
that we need to know every single person
or how video encoders work
so it's an application oriented approach
perhaps even an action-oriented approach
that cuts through
some of these questions or sidesteps or
saves them for savoring you know after
the project has been completed after
we've made the changes on the real world
it savors some of these truly excellent
philosophical and important questions
while also allowing us to act
reasonably in the moment

and so it's always that
north star of a unified model of
perception cognition and action that's
so motivating so even as some of the
formalisms do evolve

we hope that we can respect the the
spirit and the inertia of the research
that we've carried out and also know
when we can modularly swap in new
elements of the framework so blue and
then of course any questions in the live
chat

so

as a modeler

how would i model

this conversation really like let's talk
about how to really model it um and not
from a philosophical standpoint like if
this is my model of the fep i have a
model of the fep that exists in my brain
as does daniel steven maxwell we all
have a model of what the fep looks like
every time we read a paper have a
conversation take input from our it's
the action perception loop right my
model exists it's in my internal states
through my senses i perceive read here
something that either confirms my model
provides evidence for my existing model
reduces my uncertainty about my existing
model so i have some evidence input
through my senses then i update my model
or if it doesn't jive or if it makes me
realize something i mean always we're
gaining information so there's this
epistemic value add
through doing these conversations these
live streams reading papers having these

discussions etc so my my model is
updated through these conversations and
then holistically we kind of come to a
place where
our models either
converge through this communicative
process on something that resembles each
other's models more similarly or not
right we might have this divergence and
then we go away with like this
uncertainty like is my model wrong or or
what was that argument about i mean
everyone's been in conversations or
discussions or heated scientific debates
where
you don't really
come to a consensus right there is no
general agreement or understanding and
so then you walk away with maybe some
some maybe my model is invalid and then
do some more deep thought about it i
mean that that's how i would
that's how i would model it that's just
me
in the viat can at all 2020 paper with
active inference and systems engineering
systems thinking we use this
font framework formal documents ontology
narratives and tools and so one kind of
fun parallel or a way that conversation
and the real dynamics that arise can be
modeled in active inference is like
agree to disagree
two individuals can have high precision
and convergence on a narrative inference
we're agreeing to disagree
they can both be precise and agreed on
that and then at the first level they

can literally disagree
that's the whole point
so that is
modeled in active inference as
nested models just like maxwell brought
up
and so we have nested models and the
second order matrix can be
the same between two agents
yet their first level matrix can be very
different
and none of those matrices are the
system map territory
mel andrews acting 14. so we've come
back to these ideas and we'll continue
to
dance around them many many times
because they're
so fundamental that sometimes they're
hard to see
play out so steven and then anyone else
yeah i think we
have to be
careful
to note that it appears as if
um information
like what active inference tells us it
appears if
knowledge is coming in
and that it's being we we see it in our
heads so to speak and
um
yet
the this conversation is is we're we're
tuning
through the action of our info niche our
semantic niche of talking
and we're aligning

what our actions
give us
in our externality
be it through the diagrams be it through
this document and the slides or whatever
the boundary objects are that we can
align to
given us partly due to our own folk
psychology over the last 150 years the
sense and this thing is really hard to
counter this because this is kind of the
way we tend to think you know that
we have it in our minds
i so to speak
so it's true there's like this
generative model yet at the same time
the generative
modeling
is also part of what we're doing we're
generatively modeling each other's
conversation
and getting a sense of how i
should say my actions of words
none of those sentences were pre-made in
my head you know they
it's an action policy that's close
enough to my sense-making so that it
feels like i'm on the right track right
and until someone looks at me really
strange this is the um i'll carry on and
i've been in these conversations a few
times a bit more where how to
keep doing right but i think that's
something that's quite useful to when we
talk about this modeling is
oh we're modeling the action policies
that are externalized and our generative
model

is
as good as it is at being able to enact
um through participatory sense making
what should happen next
more than it is about having
something that we've received
in our generative model
thanks steven that speaks to the
interactionist versus instructionalism
like is the fep model that blue is
describing this generative model of the
fep that's
um in or embodied by all of us is it
like a skyscraper made of glass and
piping and there's 90 degree angles or
is it
some other type of metaphor that or even
transcending metaphor
and only realize through
certain sequences of interaction
so that's um really interesting
and uh cool yeah there's great comments
dave's uh throwing some biblical
references in the chat
and someone wrote i'm here first time in
the live stream it's so good to listen
all the cutting edge discussion about
fep keep up the good work so
nice work stephen and blue but thanks
also for joining you have
tuned your regime of attention
hopefully you're having some
questions and thoughts come to mind just
as they are for us we're just on the
other side of the video chat window and
the participatory sense making side
and how to mobilize that and enact it
we can have participatory sense making

with people on the video stream those
who make asynchronous edits and
improvements to these slides like we
still have to go through blues
prior thoughts that were added to um
this slide deck and then
from a history of science perspective
and sociology like it's happening in a
new way
one thing that
what was impressed on us
i guess
months to years ago forever ago in
crypto active was like the time scale of
these discussions
going from
peer review and annual conferences in
person in annual or multi-annual cycles
to like the pre-print
and then we send the twitter message to
the author of the preprint could you
give a guest stream
and then the authors can respond like
looking at what maxwell wrote which was
very important or said
like bayesian mechanics that was live
stream number 26. so we did a zero
contextualizing video respecting that
people have many different backgrounds
approaching the work from
and then two of the commentaries
beel and
um also aguilaria aguilara were in guest
stream number five and seven
so it's like we've
heard from many of the important
perspectives
and that's this participatory scientific

sense making

were all part of a bigger positive
project that's transdisciplinary it's
not even just happening in the
scientific literature

but it has to a large extent been
because these are research research
advances these are conceptual
theoretical advances
and now we're all learning by doing
and increasingly doing and applying
and so there's so much to learn and do
so um blue and then stephen and then
there's more great questions in the live
chat so go ahead

so

you're like giving shouts out to us but
i just want to like thank daniel for all
the scaffolding that he's provided in
the actin flab like to give us these um
kind of

the baseline for these discussions to
take place um and also uh just to kind
of um go back to what stephen was saying
about you know about modeling and
enacting um

really like i don't see knowledge as
coming in right like when i'm in the in
the modeling

sense like i don't know that what you
have is knowledge or what daniel has is
knowledge or what the paper contains is
knowledge um all of that in my world is
information

and i i think that there's a difference
between knowledge and information and so
like when i'm interpreting comments by
maxwell or steven or daniel when i'm

interpreting that
i take that information and
transform it through my model right like
so so some of what like take it or leave
it or take what you want from this
discussion and it's always like that i
mean we always do that whether people
have to remind us you know to do it or
not right like you take away the parts
that validate your model right
or that improve upon your model or that
you believe improve upon your model in
some way so so then when i have it and i
i have transformed it then maybe i can
turn it into knowledge but i don't see
coming in anything coming in as
knowledge that's just me
and with video chat it becomes super
real like when there's lag all of a
sudden it's like you're at the semantic
level and then it's like wait just the
syntax i'm not hearing the words and so
it's like we dip into this
lower level and then imagine if the
audio is working fine you can totally
hear
non-equilibrium steady state
but then conceptually what does that
mean so these nested and overlapping
layers of meaning so that's just very
cool so steven and then there's a
question in the chat
yeah this is a good point blue and
i suppose it's there's also this sense
that
there's always got to be some belief in
what you think the other person's saying
like for instance i believe the words

mean xyz because i know the english
language right if we're all speaking in
the language didn't know so
i believe this word means this i hear
this word these words together mean this
so there's
there's always and i'm hearing it seeing
it i'm feeling it the fact that my chair
is comfortable changes things right um
so
there is this um
this
sense that
the
you know they talk about this in free
energy principle is like it can also
appear like you know so things as well
as there being bayesian influence that
you can appear like there's a bayesian
influence type process which is
what's happening
my models can be updated and i think
what i'm really interested in is
for some of these things to then to be
capitulated as knowledge
i think there has to be and this is
where the 4e work comes in
is we we bring this into some
subconscious embodied meta well embodied
practices which we can really tap into
by sort of eliciting what they're like
maybe getting a metaphor it feels like
the knowledge is coming from around me
and i'm grabbing it some other people it
feels like it's coming in as a sequence
of very clear things
so my way of
assembling

um
the
connections of my belief structure to
the
sensory information coming in
will capitulate my modeling
and
as best i can i'm going to be then
trying to attune to the way everyone
else is and we're all i'm sure it'd be
different if we're all standing up
during all of these talks
we might actually interpret and process
a bit different if we're all in a pool
having a cocktail you know i don't know
i think blue's on a stationary bike
about half the time
i have my stationary bike hidden out of
you but let's ask a question yeah
exactly we could do a ride stream
um we could do a pull stream but you
gotta come to my house
cool i remember that so
let's ask a question from the live chat
and then we'll return to some of blue's
previously phrased questions
about the paper and that's what's kind
of fun like in the dot zero we get a lot
of the details and the bulk of the paper
and all the key ideas and figures if
there are them and then in the dot one
there's so many directions to go
and i remember earlier it's like wait
two times two hours group discussion on
a paper
i'm used to academia papers being
discussed for one hour if ever and
people having very heterogeneous

understandings or they did or didn't
read the paper so
yet
we find that there's always enough
questions when we open the doors of the
conversation to just like super
fascinating
direct directions for the discussion so
again we'll ask a great question in the
chat and then we'll kind of turn to some
of these one questions because next week
hopefully we'll have one or both of the
authors to be talking with
so joseph wrote
how has active inference
changed the way you view the world in
daily life
that's a big
question
do either of you want to go first or i'm
happy to go
i yeah i can chip in if you
well one of the ways i've actually found
it really helpful
and i thought is um
changing the way i think about how i see
the world
um
and also a bit
change that i think about playing soccer
so
uh
so the first one is
there was a there was this sense that
you know
i i i'm seeing the world right um and
it's all a bit separate from me with
active influence it it can be quite nice

to go out and think about actually
perturbing the world and feeling the
world as if you're feeling the world
with your eyes club touched touching the
world with your hands seeing the world
and touching the world with your eyes
and i think
that actually
can help bring me out of my
head so to speak in a metaphorical sense
so i found that useful actually just
thinking about that the world as being
something that's
out there and it's very much something
i'm sensing and uh so that takes a
little bit of the
modernistic kind of concreteness and
then with soccer as well
it made me realize i used to always
think you see those things on telly like
you know like someone's kicking a ball
you know like the equation's a bit like
um terminates you know
but now it's like well actually it's
it's more like i'm just
i'm just getting whatever my feeling is
right and that whole
other stuff it doesn't matter so then
you realize all these really great
sports players
the really the ones that you
they're they're actually they they've
they've acclimatized to
what it means to like methody must have
this feeling for what it means to kick
he doesn't need to have the trajectory
of the ball in his head so anyway i
don't know if that helps us go through

my crazy head
blue you want to go or i have okay so
there's a few angles and this could
definitely be
a worthy topic for another discussion
but just to give a few angles so first
thinking how we view the world that was
a visual illusion in the original
question
and so it reminds me of the
development of thought
from seeing the world that's the signal
processing paradigm photons are pouring
in and we're seeing categorizing
classifying the world okay
then we have predictive processing
and that's like well you know our visual
field
for most individuals they see color in
their entire visual field and there's
equal clarity in their visual field even
though there's an extremely
heterogeneous distribution of resolution
and color elements in the retina and we
have a big old blind spot so color
clarity of vision and blind spot
suggests that we're experiencing a
generative model a predictive model a
vision that's predictive processing and
it applies also to other sensory
modalities and then that third step was
where stephen took it with the active
vision the epistemic foraging
and we know that even though we don't
directly experience these either
there's eye circades which move to the
most informative part of the visual
scene

so to me it's about moving from signal processing 1.0 to predictive processing to active sensing and exploration as informed by epistemic and pragmatic value together you know eyes on the ball but you got to also see more than that generate more than that so that's one really important part the second one i'll also parallel exactly what steven said which is like it made me think even though i don't know about soccer i don't know about the tele wrong continent but it's kind of like the top down the video game model where there's like i see it more and more in sports replays where it's like it will pause and it will look like a video game but that's not the action model of the player on the field they're using much more of a personal flow and a feel and interactions and cues and so they don't need to be having this 30 000 foot view they need the parry personal perspective so that is one piece and then just one a third one but i think another there's many more we could list would be like the reward oriented worldview so reward maximization is the paradigm reinforcement learning is how neurons associate utility maximization is what investing

is about
in contrast with an uncertainty
reduction paradigm
a precision optimization paradigm where
qualified levels of precision of nested
models
allow a lot of flexibility
and a lot more realistic phenomenology
for different models
because precision pursuit
on an optimistic world model gets you
utility
so i just think we get utility from a
precision first worldview but from a
utility oriented worldview we don't
always get precision
and that
i personally connect to how if we have
like a tetrahedral based geometry of
nature we get cubes but if we start with
cubic geometry we never get tetrahedra
so that's sort of where i connect it
utility is like bigger and a more
general framework because utility falls
out of precision on an optimistic
generative model but merely having
utility it's always going to be ad hoc
or secondary where does curiosity and
epistemic gain and all these other
features of real decision making so
those are a few of the key pieces how it
changes how i've seen the world blue
so
besides from like having integrated
fully into like my daily vocabulary
which is like you know we talk about
things like the generative model like
you know in completely unrelated offline

conversations um so aside from like you
know enhancing my vocabulary and
building this new language of you know
the fvp um

that which is that's weird right like

okay um

but aside from that you know one of the
things one of the recurrent um themes
for me in active inference has been
um the idea like we we touched on it in
the mental action paper that we did like
daniel you're gonna have to pull the
pull it up for me because i can now
remember you've got a better track
you've got your model is updated and
configured with all of the appropriate
um

live stream numbers more more so than
mine it's 28. it's 28 and it's only
because of my extended niche but
continue that's it and and so can you
also query your extended niche because i
can't ever remember the one where we
talked about anxiety like was it if it
was big five or sophisticated active
inference or i always ask which one that
one was but i can't ever that was a long
time ago uh um but but going back you
know touching on mental action and
meditation and the idea that we touched
on way in the long ago live stream of
anxiety representing you know trying to
project too many time steps ahead into
the future like and having this relate
to anxiety and and then like depression
is relating to you know fixating too
many steps in the past behind you right
like so this like model of like

neuroatypical

things it just always reminds me to come
back to the present moment and i'm
looking for an excuse to do that anyway
just because of my inclinations toward
eastern religion and so forth so but but
the reminder to stay in the present
moment has now been like made through an
appeal to my logical brain like outside
of a spiritual
sense and so that that kind of is is a
cool you know effect of of active
inference i think

awesome

there's so many fun

questions in in the chat so i mean i'll
i'll ask these and then we can we can
we'll still have time to return to the
paper so mihail asks okay one
if you were to let your imagination roam
freely where do you
wish

or see the best possible real life
application so for that question
that's a question for
us

and the community and beyond
so we can maybe each just mention one
thought that we're having but you know
if you see

a possible future state

active inference of x

let's act infer serve

let's act first update our models and
serve and make the impact that we want
to see so this is actually an affordance
of a question because if two people are
like wow we should do this

or could this be useful
if you can state it you're like 90 of
the way there and we hope to develop the
sort of
um transdisciplinary tools
like active inference agent software and
the expertise tools people and ideas and
the education so that when somebody else
who's not on this live stream there's
just three of us
has that
insight or sees it being applied in a
new way
we want to help you scaffold that action
trajectory but blue and then stephen
so i guess i always come back to this
too something that fascinates me fully
is how information is integrated across
scales so how is the model of of my you
know dna strands imported into my cells
imported into my tissues imported into
my organs imported into my sense of self
right like so how does this and how do i
import into the organisms that i'm part
of like the active inference lab and so
forth so for me really like
concretely realizing a mathematical way
to
find information passing up and down
like is it salience or what else what
other way this emergence like for me
that's like it like if active inference
can nail that and this is like a fantasy
daydream and i know people are working
towards that but it's out there in the
future but if i could really wish one
thing if i could you know daydream about
my ultimate fantasy of active inference

being fulfilled it's completely that
okay awesome modeling nested biological
systems something that we've seen
qualitatively
discussed in the literature like
ramstead at all 2018
answering schrodinger's question
variational ecology variational
neuroethology a lot of contributors to
these works but that kind of a model
would be epic stephen
yeah i'm really interested i think
active inference
it doesn't have to do everything to be
really a contribution it can it can it
can fill certain gaps as well or give
certain mechanisms
that can give an insight either to
tractably give away to model or a way to
just think about things so
what i'm curious about and i'm working
on something called conceptual action
sociometry which is a
an action method
um i'm trying to
i'm really interested in how to
work back from a high complexity context
which would be too many dimensions to
model from first principles
so rather than modeling
the generative model per se
i'm looking at how to
create an instrument
to
track the info niche be inscribed
by
the agent so it's almost like a trace
um

so

i'm i'm

really interested in how we can

go both ways

as well as the sort of general

extractable

sort of principled work that can be show

how generative models are happening and

being applied across things like context

of autism and other areas where there's

different types of

classifications i'm really interested in

kind of contextual work which will be

outside of the ability to do full

research but might be able to do an

inquiry process and

how to sort of just get a grip on that

inquiry so

um be it people looking at their lived

experiences and things like that and

we're actually talking about that in the

tools

section people are sort of giving me

some advice

or open to that once

we get to the next phase

awesome thanks for also sharing that and

then i think the system just the one

that i'll mention would be

online teams or

research

knowledge

inquiry

and doing scholarship

and just at this point in time just

facilitating the reduction of our

uncertainty on the history of active

inference and free energy principle

literature

in a few other areas because um tools
for research and sense making would be
really awesome for a lot of different
reasons so cool and then the second
question um from the hill in that
sequence was

what if we model our group conversation
where we as humans disappear
and where we are not exploring the
universe but the universe is exploring
itself utilizing us

a beautiful

thought question and there's a lot to
think about that so blue

yeah i mean i think that this uh

is

possible right and

you know

we've talked about it in the
computational boundary of the self the
mike levin paper like
ultimately it's like dissolution of the
cognitive boundary and is there some
like meta organism this goes to like the
gaia hypothesis maybe not the universe
but the earth or maybe the earth is part
of the universe maybe there's some ultra
ultimate like super organism that that
cognitive boundary contains all of us
and so just in the same way that like
our cells are working towards our
optimal function given the circumstances
that they're they're presented with
right like given that assumption maybe
we're also working toward the optimal
state of the universe like every
interaction that occurs between all of

us is working toward the optimal state
of the active inference lab or the
optimal state of the united states of
america or the optimal state of you know
scientific research or who knows right
so any kind of
you know

umbrella category that we might fall
into maybe we're serving the optimal
state of that and really like i think
it's

quite possible and a really cool way to
model active inference for sure
another

take on that from bucky fuller is this
function

first mentality led him to ask well what
is the function
of human

and are we local sensemaker what is our
role so those are big important
questions

let's return to

31.1

and kind of prime ourselves for our dot
two because in the dot zero that's where
like whoever's really excited about the
paper maybe whoever contacted the
authors or maybe one of the authors
themselves or just a few people who want
to learn by doing the dot zero which was
like blue and i for this one 31.1 the
previous video

we go through the paper the dot one we
unpack that's sort of the bathtub phase
everything is getting spread out a lot
of sociological
criticisms or uh insights a few random

questions from the get from the guests
of our stream
a lot of different areas a lot of
threads get pulled out and then in the
dot 2 we try to have some synthesis
because it's not the end of the
discussion with the dot 2 but
we have to move on from a live stream
perspective so the dot 2 especially when
the authors can be there we can get some
more insights we can resolve some
questions that arose during the dot one
and dot zero and the inter
uh meaning times so questions that
didn't arise
before we enacted
this process
and then we get to ask those and then
kind of use that as our jumping off
point as we move on to our next phase so
here in the dot one
at the end
it's kind of like a fractal.2 we're kind
of trying to tie a few things together
returning to the text as a strange
attractor so that we'll have a whole
week to read think sleep
stationary bicycle
mind and body relax
maybe do our day job if we can find any
and um
be here you know next week or whenever
works to continue the discussion so blue
um in this questions for 31.1
what were some
quotes or some questions from the paper
that you wanted to like learn more about
or ask the authors about

well just as i mentioned um you know
even earlier in this live stream today
like
the dynamic
equilibrium of living systems like i'm
not understanding that at all like maybe
that's homeostasis but homeostasis is
not in my mind dynamic equilibrium and i
think that this um you know bayesian
mechanics for stationary processes and
the idea of a non-equilibrium steady
state density
uh kind of supersedes
this concept of dynamic equilibrium
because homeostasis is not equilibrium
and so that's kind of a fundamental like
it like bends in my brain a little bit
when i when i've been reading over this
paper um
and also uh you know it says here we
should pay closer attention to
fundamental differences between physics
and biology um and how these
indifferences interact with importing
concepts from physics into the modeling
of biological targets so
i mean
is it physics all the way down because i
always thought it was physics all the
way down right like so i don't know that
goes
to um
you know like
when i was in school they always said
like you know biology is not as precise
as chemistry and chemistry is not as
precise as physics and physics is not as
precise as math right like so there's

this like tiered hierarchy of how you
know fundamental your
study can be right
and so
it i think biology is based in the
concept of physics but but also
chemistry i feel like was really omitted
and left out and like in my heart i'm a
chemist uh stephen i'm sure can relate
to that
um but really like so so that's kind of
one other uh thing i wanted to explore
and then um
the other part of um
of the questions that i have for the
authors are you know in active inference
are the active states internal um
because they said in internal states
include active states but i have always
thought of active states and sensory
states as part of the interface between
internal and external states as part of
the markov blanket and so i just wanted
to clear that up maybe if that's a
misunderstanding on my part or
i mean i guess action can happen
internally like as we've seen in the
mental action
so that's definitely possible but but
not all active states are part of
internal states
i thought so i don't know maybe that's
some something that we can tease apart
with the authors and i'm looking forward
to having them here and asking them
these three questions which are the ones
that kind of for me
um you know don't align with my current

model of the fbp
they were salient they
were different from your generative mall
so it's a nice way to put it and then
i'll just show this one comic and then
steven so this is a classical uh xkcd
comic
the field's arranged by purity okay so
it's a comic you know
retweets are not endorsements but you
have
uh this is a classic disciplinary
hierarchy slash
ponzi scheme pyramid scheme whatever you
know
you have uh the physicist let's just
start in the middle it's nice to be on
top that's the the common
pinnacle of the pyramid in certain
paradigms and then like the physicists
can make claims about lower levels but
then if the psychologist makes a claim
about physics hey we need a sociology of
molecules it'd be like that's
not science
but then the physicist talking about
sociophysics is like wow that's so
modern and exciting so there's a sort of
um linearity
as well as potentially some sociological
scaffolding that facilitates the
diffusion of information and expertise
and uh notoriety like one way but it's
not always a two-way street and then the
joke i guess in the xkcd is like the
mathematician saying hey i didn't see
you all from over here so again just a
comic now

playing with the media what does it look
like to i don't know close the loop
or
have these
learn about each other's perspective or
there's a million ways that we could
go counterfactual on this xkcd
what if i mean that's the name of the
xkcd's author's book which is kind of
fun
um but how does active inference
challenge this
either specifically or just how are
broader ongoing sequences of actions
like
decentralized science desai open science
open access open source global
collaboration
transdisciplinary application complexity
science i mean there's a million ways in
which this is kind of funny because it
speaks to
an academic hierarchy and maybe we've
even met some characters who really
embodied these archetypes but
what are the archetypes going to be
going forward what are more
accessible
archetypes
and ones that are more true to form and
they can still be funny i'm sure we'll
have acting memes just enough to warm
the hearth
but this is just one view that kind of
speaks that disciplinary hierarchy so
yeah steven then blue
yeah i think this this is where this
foundational shift is this this desire

for purity

um starts has been really challenging of
course now we're in this post-normal
science era

where you know the reality is that we're
gonna have to make decisions and not
just wait for the purity of the
the random control trial or whatever it
is to make choices so how can we be more
um

systematic and rigorous and so then
complexity comes in and
the challenges that we are facing is
that a lot of psychologists and
psychiatrists um have become i've also
gone down the purity route in a way
they're trying to like the idea is
they're all trying to be more pure and
the mathematician can be more pure than
the others but that question about being
more pure

i think this is what blues

points make a really good

you know

they come into question like physics
comes into question chemistry comes into
question

when you talk about

far from equilibrium states

even the idea of it and in chemistry

they call it far from equilibrium

chemistry in physics they call it

non-equilibrium states right

um because they can actually method and

the mathematicians would call it lorenz

attracts estates right and but in either

case you know you only have to look at

lorenz attractor the whole point is you

don't know where you are in it right and
and not in the same way as quantum
well that's the question whether you
when you get into the fine details we
don't know yet but it's still different
to quantum right and i think
that's where that quantum contextuality
comes in as well so anyway what's your
thoughts on that
the thing that just came to mind and
then blue is like this is a linear right
it's open-ended um but
sociologists
and philosophers make important insights
about the philosophy and history of
science
and so the mathematician can say yeah
we're the most pure or we're the most
correct or whatever it happens to be but
they're nested in a sociological context
in a historical in a philosophical
context so
it is closed from a causal closure
perspective because it's not like
physics have just broken out of
the frameworks of thought
so
how to move this line
into
a conversation that's
including all of these disciplines
in a trans discipline and
including people who aren't in academia
or research in that as well blue
so
this is an interesting comic and i i
usually like don't even talk about
psychologists or sociologists because

that's not science right no i'm just
kidding um i really actually have a lot
of respect for psychology and sociology
um but it's just not something that's
ever been like lumped into my pyramid
before so uh thank you for for sharing
that and seeing it in this way like
maybe

they should arrange by like the
fundamentality of it like sociology
without like social interaction

we would not

develop things like mathematics right
like there would be no need so so in
terms of what's more fundamental like
maybe sociology is fundamental and
psychology like we're not even thinking
about what makes us work if we're not
psychologically

okay right like if we're in a bad
psychological state nobody's thinking
about intra cells and dna like that's
there there's just no you have to have
like the fundamental needs have to be
met before you start to ponder things
like what am i made of and how do my
parts work right and and really like so
so maybe in terms of fundamentality
and then we are biology and and then you
know so maybe in terms of the
fundamental things sociology is the
pinnacle of that

and

again there's probably many ways in
which different perspectives or active
inference can

challenge this or at least kind of
continue this joke a little bit but it's

like

pure active inference it's really about
sociological context we really have a
mathematical formalism it's really
grounded in the ways that people have
been studying physics that's one of the
questions of 31 the paper is to what
extent is it has it could it been
grounded in physics but
we really have physics
and we really are talking about biology
and for comprehensiveness and and
getting to the fundamentals it is really
fundamentally relational
so we're not taking this reductionist
purity spiral we're taking like a
transdisciplinarity
purity and we're just saying this is our
keyword
and to really get out what we're talking
about we're really going to respect all
of these perspectives so that's kind of
a funny challenge that you wouldn't
necessarily hear
arising from a disciplinary conversation
so stephen go for it
yeah i think this is also one thing you
know we say though that
we see the importance of social
interaction social inference and um
the challenges that i think actually
this might be quite apt for where we had
at the beginning with maxwell's
conversation is
the perspective of the sociologist
becomes what is ascribed to social
interactions you know the psychologists
is what's ascribing the mental models

that we have in our head the biologists
are ascribing the categories the chemist
is describing the steady state you know
so
when you get into this agent-based
modeling which again i think is is
more familiar to the newer work in
biology but it's
something that i had to start to learn
about
you know and looking at these dynamics
and modeling
actually
um
starts to
it starts to go under the ground of all
of these
in it doesn't make them invalid because
they they are useful when you can take a
perspective on the systemic work
but
we're not taking perspectives in this
and there's that's the whole ground
starts to become
shifting um and that's really
interesting
okay here's another
kind of a subtlety actually so
um sociologists okay just starting there
that's our prior you know d coming in
from the left sociology is applied
psychology okay so we're going all the
way you know
chemistry is applied physics
it's just like
it's interesting the conversation is
going that way
i'm just seeing another comic where it

could have been like
physics is applied math
chemistry is applied physics like who is
saying
which field is an application of which
other fields so there's a lot of ways
that conversation could play out
and by imagining the counterfactuals
the semantic the rhetorical
counterfactuals
and then asking well what is concordant
or discordant with our generative
model and why
that's a very rich starting point for
transdisciplinary discussions because
the whole
inter meta transdisciplinary discussion
it's been decades
centuries truly with different words
so how are we going to
look at this with a new
light
and maybe even integrate some of the
truly recent developments in tools
like performance computing
and interactive notebooks and ontologies
these didn't exist one or two hundred
years ago in the same way they do today
so how are we going to
respect qualitative and quantitative
and different backgrounds recognizing
that we're never going to get
the end of the story even if you got 100
phds in any one of these fields or lived
100 years in a field outside of academia
you still wouldn't be at the end so
how are we going to take a radically
different approach

to
learning and applying
maybe active inference could have
something to do with it
so
in the last few minutes
anyone else in the live chat could just
ask a final question otherwise let's
just um sort of recap the big
aims and claims of this paper
were about
the foundations in statistical physics
and dynamical systems theory for the
free energy principle
and what's on
the line what's on the table for whether
the theoretical underpinnings are
adequate or not for fep and active well
it's the practical utility as well as
the theoretical adequacy
so
that's what matters
they claim that their work is starting
to fill in this gap
and whether we think that there's any
any other pebbles in that gap or the gap
is uh you know a meter across or a
kilometer across there is a differential
and we can all positively contribute to
that through
discourse and through contributions to
the literature
and then the main argument of the paper
and one of the big ideas that'll be
really cool to talk about
is that the foundations in physics allow
the fep models
to be maximally general to describe

across systems
but in the author's words these
assumptions also decrease the ability of
fep models to include enough factors to
provide biologically plausible
representations for living systems and
the way that they are in the world
so you know big idea
physical and physics grounding of fep
and what that says about the theoretical
adequacy as well as the practical
utility and then this idea of
generalization
and if we over generalize do we sort of
leave behind some of the features that
we wanted to explain in the first place
so that's a little bit of the recap
and i'll just
pull to our kind of closing slide and
ask these questions and then
stephen and blue you can give a last
thought
what would a good
understanding enable
what are the unique predictions and the
implications of this research or of what
you thought of after learning about this
research
what are the next steps for fep and
active inference
what are the goals of this or your
research and then what are you still
curious about what is your intrinsic
motivation that's going to keep you
reading these papers and contributing to
the discussion
so stephen or blue any last thoughts
stephen first and then

i think

in terms of this paper anyway i think

that this really does highlight this

need to understand the difference

between particles

voxels things systems i think this

because these types of critiques keep

coming up right at times i know it's

it has a risk of kind of tying

the free energy principle community in

some knots um and uh

so i think that

it's a challenge and i think that that

that may be a part of where this

goes and

at the same time i think it's a good

provocation and

there is some questions here

about how and why those confusions arise

and

so i think that might be uh um you know

i'm curious about

learning more about that because i think

that's going to help in other ways in

doing this work

thank you stephen blue

so

it's awesome to me i think that i'm

talking about the

um general general generalizability

sorry that didn't come out well the

generalizability and applicability of

like the the

free energy principle i i really like

that that's also like a trade-off

between like epistemic and pragmatic

value like it reminds me of that balance

that we see in the fep and and so i

think like the ultimate goal is to
resolve that right like how do we strike
the balance between epistemic and
pragmatic value how do we strike that
balance between generalization and
application
great point we want to describe many
systems and describe them well and walk
in with a scaffold but minimal
constraining priors that prevent us from
seeing some fundamental or even
essential attributes of systems and then
that general in specific
connecting that to learning and doing to
epistemic and pragmatic values or gains
which
hint we have included in the actin
formalism as part of our policy
selection approach
so stephen and blue plus everybody who
asked truly excellent questions in the
live stream
will see you in an organizational unit
meeting or the discord or
some other way just
this is your affordance to participate
and contribute so thanks again to
everybody and we'll see you next time
bye